UNIVERSITY OF THE WESTERN CAPE

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Evaluating DINOv3 Embeddings for Anomaly Detection in MeerKAT Radio Galaxy Surveys Using Astronomaly: Protege

A research report submitted in partial fulfilment of the requirements for the
Bachelor of Science (Honours) in Astrophysics

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Abstract

Visual inspection cannot keep pace with the number of radio sources in the MeerKAT Galaxy Cluster Legacy Survey. Earlier Astronomaly: Protege work used Bootstrap Your Own Latent (BYOL) features, but had not compared them with DINOv3 features in the same ranking procedure. This study made that comparison using 6,161 source cutouts, of which 1,031 had human-interest scores for evaluation. Each feature set was reduced by principal component analysis and passed through the same Protege procedure. The pretrained DINOv3 vision transformer reached an area under the discovery curve (AUC) of 0.928, compared with 0.784 for BYOL. Among the first 100 recommendations, it recovered 65 of the 86 sources scored 4 or 5, compared with 28 for BYOL. Only three sources appeared in both Top-20 lists. For this labelled subset, DINOv3 prioritised high-interest sources more effectively. As Protege selected sources for training, their previously recorded human scores were supplied to it. Because the final rankings were also assessed on that labelled subset, these results are not an independent test on unseen sources.

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I used OpenAI's ChatGPT and Codex to help find original references for equations in the theory chapter, troubleshoot errors in parts of the project code, and prepare and revise the LaTeX format of the report. The analysis, interpretation and conclusions remain my responsibility.

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1 Introduction

1.1 Background and motivation

The amount of data collected by modern radio surveys has made the search for unusual sources increasingly difficult. MeerKAT observations reveal compact radio sources together with jets, lobes and faint extended emission. These structures can provide information about active galactic nuclei, galaxy evolution and the environments in which galaxies are found. However, the number of detections is too large for all of them to be inspected individually. The image representation used by a search method can affect which sources it treats as similar [Lochner and Rudnick, 2025].

The MeerKAT Galaxy Cluster Legacy Survey (MGCLS) observed 115 galaxy clusters between 900 and 1670 MHz. Its catalogue includes approximately 626,000 compact source detections, in addition to a range of extended and diffuse radio structures [Knowles et al., 2022]. A selection of MGCLS imaging capabilities is shown in fig. 1.1. The full-resolution, convolved and diffuse-emission views illustrate why the appearance of a source can depend on how the radio image is processed.

Anomaly detection can help prioritise images, but statistical outliers include artefacts and noise. A scientifically interesting source need not be the strongest outlier [Lochner and Bassett, 2021].

Astronomaly incorporates researcher feedback into the search [Lochner and Bassett, 2021]. Its extension, Astronomaly: Protege, uses those scores to recommend further sources for inspection [Lochner and Rudnick, 2025].

Before Protege can compare images, each cutout is represented by a numerical feature vector, or *embedding*. Previous MGCLS applications used Bootstrap Your Own Latent (BYOL), a self-supervised learning method, to generate these embeddings [Grill et al., 2020, Lochner and Rudnick, 2025]. In this project, BYOL was used as the baseline against which the pretrained DINOv3 representation was assessed.

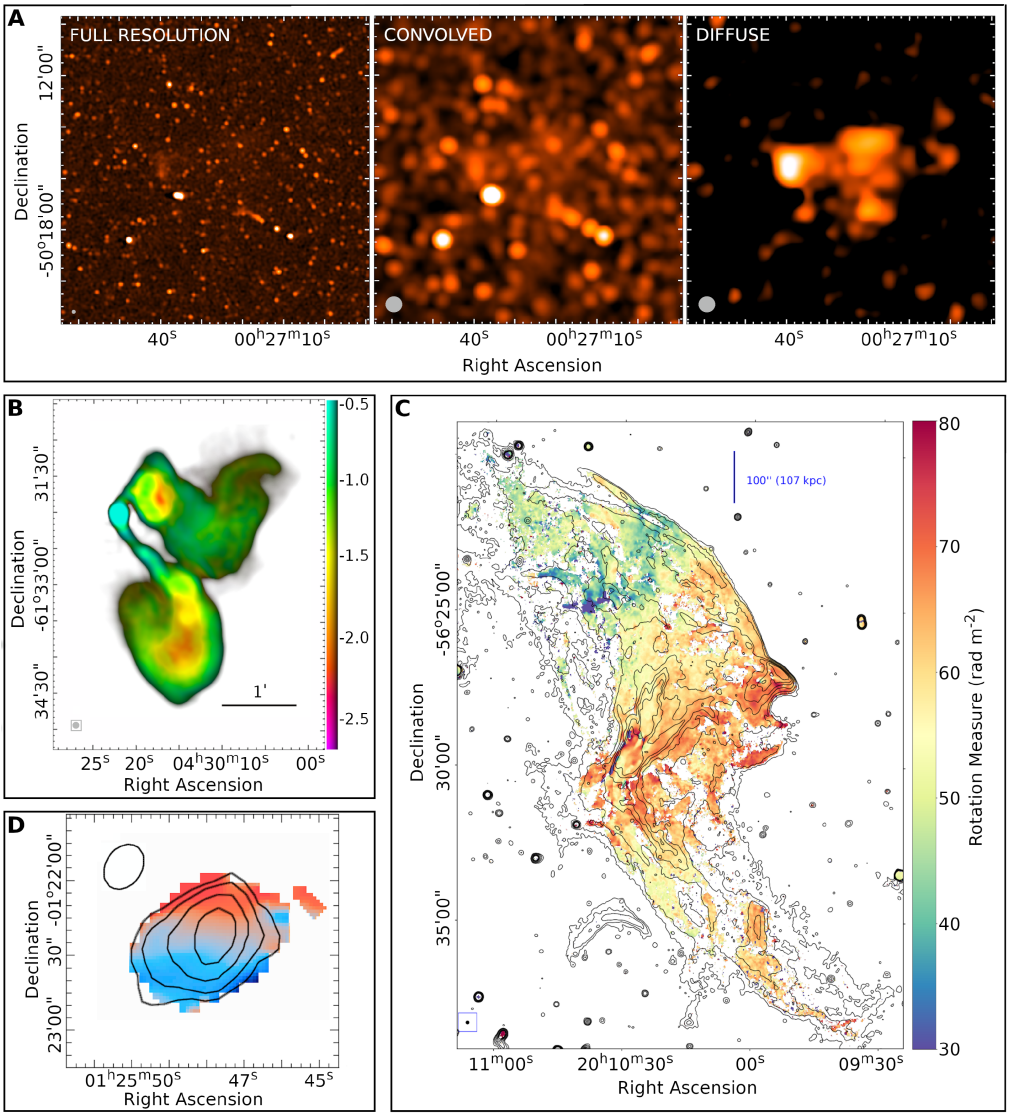


Figure 1.1: Different views of MGCLS observations, including full-resolution, convolved and diffuse radio emission (panel A), together with additional scientific examples (panels B–D). Reproduced from Figure 1 of Knowles et al. [2022]. Original panels, colour scales and labels are retained.

1.2 Previous work

1.2.1 Anomaly detection and human feedback

The original Astronomaly framework lets a researcher score candidate sources while the model updates its recommendations [Lochner and Bassett, 2021]. This is useful when the aim is to find sources that are interesting to inspect, rather than to assign every image to a predefined class.

Protege extends that approach by using a Gaussian Process model to predict human-interest scores and an acquisition procedure to choose further examples for scoring [Lochner and Rudnick, 2025]. Its active-learning process is illustrated in fig. 1.2.

The MGCLS application showed that Protege could return morphologically diverse sources, including objects that researchers had initially overlooked. Figure 1.3 reproduces four published examples assigned high Protege scores despite lower initial human scores. The examples illustrate why a ranked recommendation list can complement manual inspection [Lochner and Rudnick, 2025].

Subsequent work used Protege to prioritise diffuse radio emission in MGCLS [Etsebeth et al., 2026].

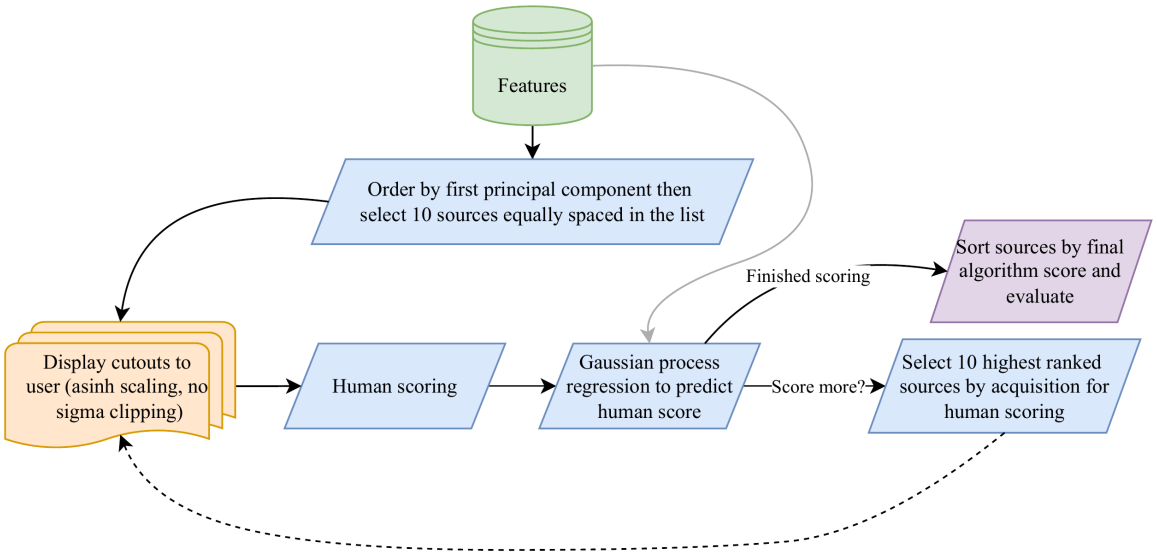


Figure 1.2: The published Protege active-learning workflow, including initial source selection, human scoring, Gaussian Process prediction and revised recommendations. Reproduced from Lochner and Rudnick [2025].

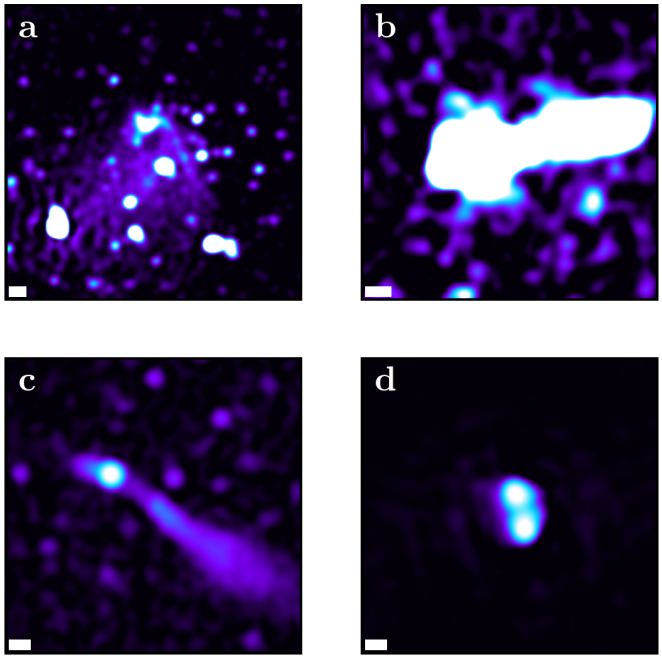


Figure 1.3: Four MGCLS sources (a–d) which received high Protege scores but were initially scored low by human reviewers. The authors concluded that sources a, b and c deserved further consideration after reinspection. Reproduced from Lochner and Rudnick [2025]. The image stretch is as published.

1.2.2 BYOL and DINOv3 feature representations

BYOL learns image representations by comparing two differently augmented views of the same image. An online network predicts the representation produced by a target network, whose weights follow the online network through an exponential moving average. This training does not require labelled images or explicit negative pairs [Grill et al., 2020]. Earlier MGCLS Protege work used BYOL embeddings to represent radio-source cutouts before the human-guided ranking stage [Lochner and Rudnick, 2025]. Those completed BYOL features form the baseline in this project. The original training run was not repeated here.

DINOv3 produces transferable image representations, including global features and spatial features for image regions [Siméoni et al., 2025]. Figure 1.4 shows spatial features from the original paper at different image resolutions. The MGCLS comparison used the frozen ViT-H+/16 model's global feature vector for each radio cutout, not the spatial maps shown in that figure.

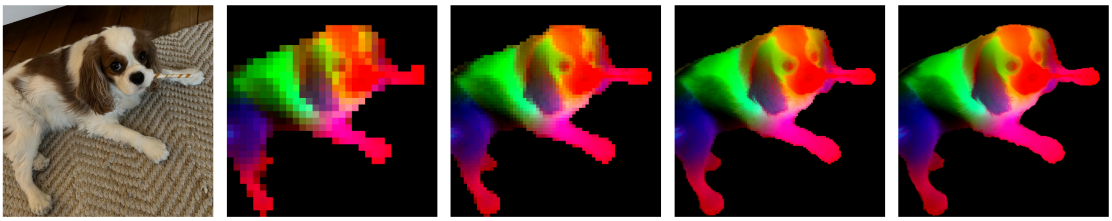


Figure 1.4: DINOv3 spatial features visualised at increasing image resolutions. The coloured maps show principal components of dense features from the original paper. They are not radio images or the global embeddings evaluated in this report. Image panels cropped from Figure 4 of Siméoni et al. [2025].

The BYOL features in the MGCLS Protege study were learned from radio cutouts [Lochner and Rudnick, 2025]. DINOv3 supplies pretrained features without a new MGCLS training run. Its published results on other visual tasks do not show how well its global vector represents faint radio emission [Siméoni et al., 2025]. This study compares the rankings produced by these two sets of features.

1.3 Research gap and aim

Previous Protege studies show that BYOL embeddings can support human-guided source discovery in MGCLS [Lochner and Rudnick, 2025, Etsebeth et al., 2026]. Neither compares a pretrained DINOv3 representation with BYOL under a shared Protege evaluation.

The aim was to evaluate DINOv3 embeddings for anomaly detection in MGCLS radio-source images and compare their ranking performance with the existing BYOL implementation. The study used the same working catalogue and the same labelled evaluation subset for both representations.

1.4 Objectives and research questions

The objectives were to implement pretrained DINOv3 feature extraction, integrate those embeddings into the existing Protege workflow, compare the resulting rankings against BYOL and inspect the highest-ranked images.

> How does replacing BYOL with DINOv3 embeddings affect the ranking of MGCLS radio sources within Astronomaly: Protege?

The visual comparison asks whether the two representations bring different sources into their highest-ranked lists.

Chapter 2 introduces the feature representations and the ranking model. Chapter 3 describes the data and comparison procedure, Chapter 4 presents the rankings and radio cutouts, and Chapters 5–7 discuss their interpretation, conclusions and next steps.

2 Theory and theoretical framework

2.1 Radio-source images and numerical representations

A radio-source cutout is a sampled intensity distribution. It may show compact components, jets, lobes or extended emission, but it does not measure physical properties such as magnetic fields directly. A feature extractor maps the processed pixels to a vector for downstream ranking [Grill et al., 2020, Siméoni et al., 2025].

For an input image I_i, a feature extractor f_θ produces the embedding

$$\mathbf{x}_i = f_\theta(I_i), \qquad \mathbf{x}_i \in \mathbb{R}^d, \tag{2.1}$$

In eq. (2.1), θ represents the model parameters and d the embedding dimension.

2.2 Self-supervised visual representations

Self-supervised learning draws a training signal from images without requiring a human-interest label for each one. BYOL learns from augmented views [Grill et al., 2020]. DINOv3 is also self-supervised, but its training data and procedure differ from those of the BYOL baseline [Siméoni et al., 2025, Lochner and Rudnick, 2025]. Protege learns interest scores from researcher feedback after feature extraction [Lochner and Rudnick, 2025].

2.3 DINOv3 as a vision foundation model

A foundation model is not built for only one problem. It is first trained on a large, varied dataset and can then be reused for other tasks, either without changing its weights or after further training [Bommasani et al., 2021]. DINOv3 is a foundation model for images. It learned visual features through self-supervised training without a human class label for each training image [Siméoni et al., 2025]. This project tests whether those features help rank MGCLS radio sources. Broad pretraining does not guarantee that they capture faint radio structures.

DINOv3 provides a global representation of an image as well as features for individual image regions [Siméoni et al., 2025]. In this project, the pretrained ViT-H+/16 model was kept fixed and its global representation was used as one feature vector per MGCLS cutout. Protege then used the human-interest scores to rank sources. DINOv3 itself did not assign those scores.

The version used here is a Vision Transformer (ViT). A ViT divides an image into patches and processes their representations with positional information [Dosovitskiy et al., 2021]. In ViT-H+/16, the patch size is 16×16 pixels [Siméoni et al., 2025]. Self-attention lets patches exchange information across the image.

For query, key and value matrices Q, K and V, the standard scaled dot-product attention operation introduced by Vaswani et al. [2017] is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\mathsf{T}}}{\sqrt{d_k}}\right)V, \tag{2.2}$$

In eq. (2.2), d_k is the dimension of the key vectors. The attention weights allow information from one patch to influence the representation of another.

2.4 Principal component analysis

Principal component analysis (PCA) reduces the number of variables used to describe a dataset while keeping the main patterns of variation [Jolliffe, 2002]. It combines the original variables to make new axes, called principal components. The first axis captures the most variation. Each following axis captures as much of the remaining variation as possible while being perpendicular to the earlier axes. Keeping only the first few components gives a simpler description of the data.

For a data matrix X and the first k principal directions W_k, the conventional transform is

$$Z = (X - \boldsymbol{\mu})W_k. \tag{2.3}$$

In eq. (2.3), Z contains the reduced representations and $\boldsymbol{\mu}$ is the vector of variable means. The fraction of the original variation captured by the first k components is [Jolliffe, 2002]

$$V_k = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{d} \lambda_j}, \tag{2.4}$$

where λ_j is the variation captured by component j and d is the total number of components. Keeping more components retains more variation but reduces the data less.

2.5 Gaussian Process active learning

Protege uses labelled examples in feature space to predict interest in unlabelled sources [Lochner and Rudnick, 2025]. A Gaussian Process (GP) assigns a distribution over possible score functions [Rasmussen and Williams, 2006],

$$f(\mathbf{z}) \sim \mathcal{GP}(m(\mathbf{z}),\, k(\mathbf{z}, \mathbf{z}'))\,, \tag{2.5}$$

In eq. (2.5), $\mathbf{z}$ is a reduced feature vector, while m and k are the mean function and covariance kernel. Protege uses a Matérn kernel with a white-noise term and selects further sources for scoring through expected improvement [Lochner and Rudnick, 2025].

3 Methodology

3.1 Dataset and evaluation subset

The working dataset consisted of 6,161 MGCLS radio-source cutouts. These source identifiers were kept consistent between the BYOL and DINOv3 feature tables. The working sample is smaller than the full MGCLS compact-source catalogue described by Knowles et al. [2022].

The evaluation subset came from Lochner and Rudnick [2025]. It drew on Abell 209 and Abell S295, for which extended-source catalogues were available, and 18 other randomly selected MGCLS fields. After source extraction and preprocessing, it contained 1,031 cutouts. Michelle Lochner scored all 1,031 sources from 1 to 5 according to scientific interest. Lawrence Rudnick separately scored a random sample of 100 sources to assess agreement.

The paper gives unresolved sources and artefacts as typical score-1 examples. Its score-5 examples include X-shaped, turbulent and unfamiliar structures. These scores record a researcher's interest in a source, not its membership in a physical class. The existing scores allowed BYOL and DINOv3 to be compared on the same source identifiers.

Sources marked with a score of -1 were excluded. Scores of 4 or 5 were treated as positive high-interest labels, while scores of 1–3 were treated as negative labels for the ranking metrics. Table 3.1 shows the label distribution.

Table 3.1: Human-interest scores in the labelled MGCLS evaluation subset.

Human-interest score	Number of sources
1	384
2	439
3	122
4	62
5	24
Total	1,031

There were 86 positive examples under this definition. Source identifiers were used to join the reduced feature matrices to the evaluation labels. The two methods were then compared on the same 1,031 sources.

3.2 Image preprocessing

For this study, image preprocessing followed the BYOL baseline workflow supplied by Professor Michelle Lochner. Both notebooks used the same Astronomaly cutout settings and transformations. Catalogue-centred square cutouts were resized to 128×128 pixels without anti-aliasing,

then sigma-clipped, with negative values removed before scaling. The single-channel images were repeated across three channels before DINOv3 feature extraction.

The fixed size and central-source selection followed the earlier MGCLS workflow, which sought to reduce differences in source size and unrelated background emission [Lochner and Rudnick, 2025]. Scaling gave a common input range, while three-channel conversion met the pretrained model's input format. These preprocessing settings were kept unchanged for the DINOv3 comparison. Clipping and resizing may still suppress faint or extended structure.

The DINOv3 input was subsequently normalised using the channel means and standard deviations shown below.

$$I'_c = \frac{I_c - \mu_c}{\sigma_c}, \qquad \boldsymbol{\mu} = (0.485, 0.456, 0.406), \quad \boldsymbol{\sigma} = (0.229, 0.224, 0.225). \tag{3.1}$$

In eq. (3.1), c indexes the three repeated channels and I_c is the intensity in channel c. The mean μ_c and standard deviation σ_c are an input convention of the pretrained DINOv3 model. They are not estimated from the radio-source sample.

3.3 Feature extraction and pipeline design

The BYOL baseline features came from a completed training run supplied by Professor Michelle Lochner. This project integrated DINOv3 into the existing workflow and compared the two representations. The DINOv3 branch used the official pretrained ViT-H+/16 checkpoint `dinov3_vith16plus_pretrain_lvd1689m-7c1da9a5.pth`. The model was placed in evaluation mode and run without gradient computation. It was not fine-tuned on MGCLS. Its global image representation was used rather than an average of patch embeddings. The evaluation used the saved BYOL and DINOv3 feature tables rather than rerunning feature extraction.

The DINOv3 extraction step is summarised in fig. 3.1.

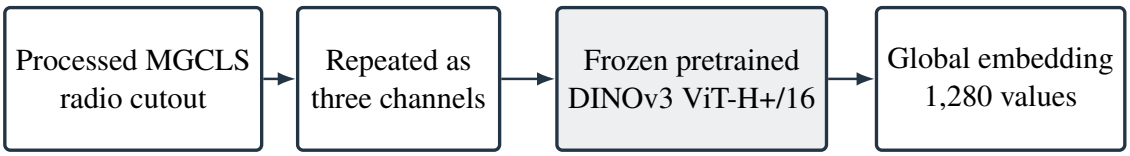


Figure 3.1: The feature-extraction step used for MGCLS cutouts. The pretrained ViT-H+/16 encoder produced one global embedding per source without further training. Diagram prepared for this study using the model described by Siméoni et al. [2025].

Each model produced a matrix with 6,161 rows and 1,280 columns. The matrices were saved separately as Parquet files. Figure 3.2 shows how both sets of features entered the same downstream procedure.

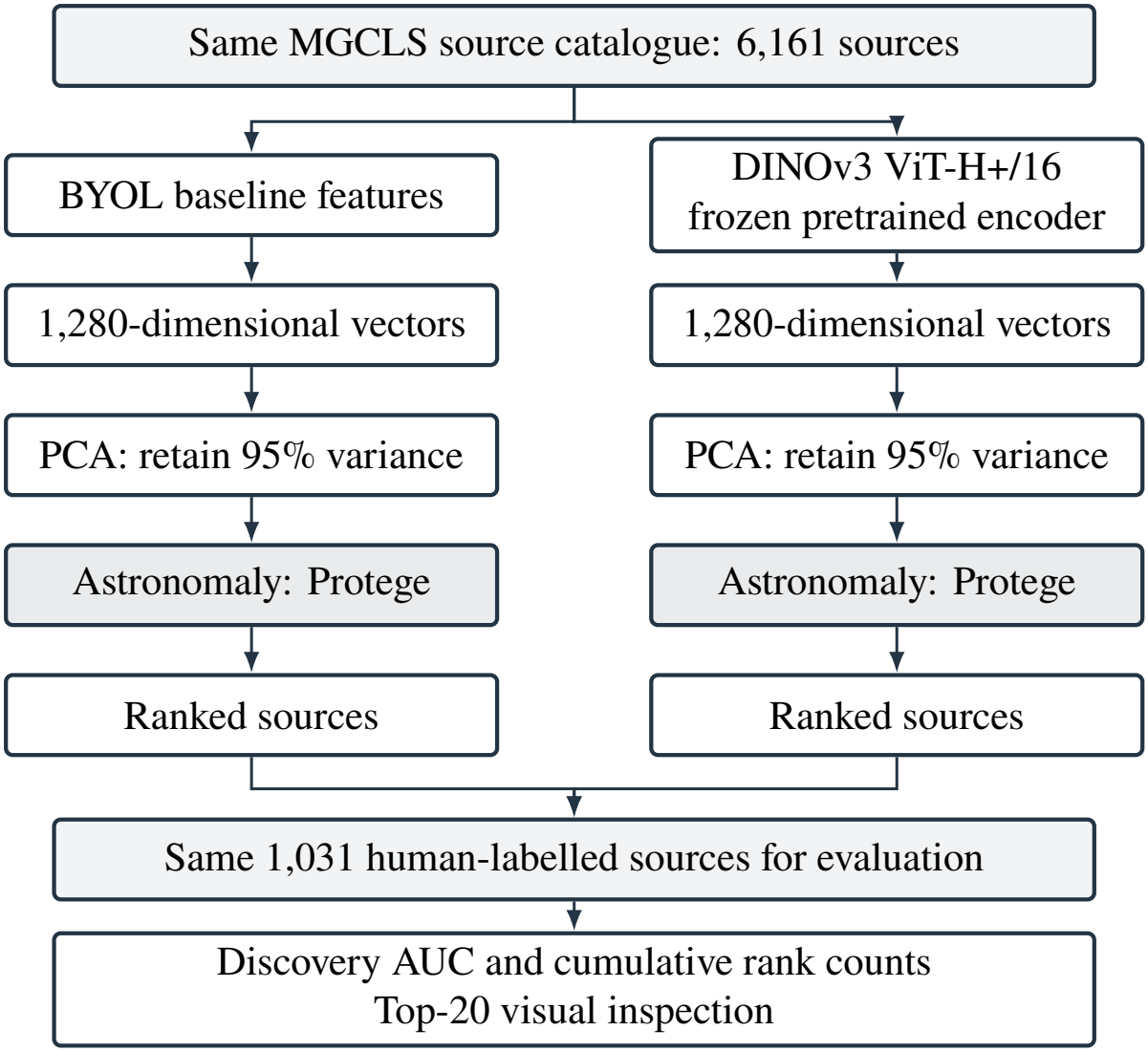


Figure 3.2: Overview of the processing and evaluation used for BYOL and DINOv3. Both models represented the same source catalogue. Each feature matrix was reduced using PCA before the Protege evaluation was applied to the common labelled subset. Diagram prepared for this study.

3.4 Dimensionality reduction

The Astronomaly PCA decomposer was applied separately to each feature table. Components were retained until the cumulative explained-variance fraction in eq. (2.4) reached at least 0.95. BYOL retained 45 components, explaining 0.9500 of its feature variance. DINOv3 retained 195, explaining 0.9502. The same variance criterion was used for both methods rather than fixing the number of components.

3.5 Active learning with Protege

The GP active-learning procedure used the existing Protege code. The reduced feature table was sorted by its first principal component, and ten approximately equally spaced sources formed the initial labelled sample. Protege then selected ten additional sources per step using its acquisition ranking, up to a label budget of 600. The catalogue already contained Michelle Lochner's scores. When Protege selected a source, the notebook supplied its recorded score and updated the GP after each batch. No new human scores were collected during these runs.

The relevant experimental settings were `initial_steps=10`, `steps=10`, `N_labels=600`, and `ei_tradeoff=0`. After the final update, the output was ranked in descending order of `trained_score`. The same label definition and comparison protocol were applied to BYOL and DINOv3.

3.6 Evaluation metrics

For each final Protege ranking, the 1,031 labelled sources were ordered by predicted interest score. A discovery curve plots the fraction of the ranked list inspected against the fraction of known high-interest sources found. This is cumulative recall along the ordered list [Manning et al., 2008]. The area under the discovery curve (discovery AUC) summarises how early those sources were recovered across the ranking, while counts at fixed ranks describe particular inspection budgets. Discovery AUC is not classification accuracy or the area under a receiver operating characteristic curve.

The binary labels defined above were used to calculate the discovery curve. Its area was calculated with NumPy's `trapezoid` function, which applies the standard trapezoidal rule [Burden and Faires, 2010]. The notebook counts the highest-ranked source at horizontal coordinate zero, and the reported AUC values retain that indexing. Counts at ranks 25, 50 and 100 came from the same cumulative discovery arrays.

For qualitative comparison, the top 20 sources from each ranking were used to retrieve radio-image cutouts from the original MGCLS Flexible Image Transport System (FITS) images. The final panels used a display-only inverse hyperbolic sine (asinh) stretch. Positive pixel values were divided by their 99.9th percentile, limited to the interval from zero to one, and mapped with a stretch factor of five. This made faint emission easier to see without changing the feature inputs or the rankings. Six sources present in only one Top-20 list were chosen for larger views

after inspecting the saved panels. Their images were cropped from those same asinh-stretched panels. This was an illustrative selection, not a systematic morphology sample. Each cutout was stretched separately and the catalogue-centred cutout size could vary, so apparent brightness and angular extent were not compared between sources.

3.7 Reproducibility and limitations

The work used Python, PyTorch, pandas, NumPy, scikit-learn, Matplotlib and Astronomaly. Complete package versions, hardware details and random seeds were not saved, so they cannot be reported reliably. The feature tables, discovery arrays and Top-20 tables are the available records of the comparison. Both notebooks wrote PCA output to the same folder. Only the DINOv3 PCA state remains there, although the BYOL component count is recorded in the notebook output.

The final rankings were assessed on the full 1,031-source labelled subset, including sources whose scores had been supplied to Protege during training. Discovery AUC and the cumulative counts are therefore not results from an independent held-out test. The labelled subset is also only a fraction of the MGCLS working catalogue.

4 Results

4.1 Quantitative comparison

Each feature set produced a Protege ranking of the same 1,031 scored sources. Of these, 86 had human-interest scores of 4 or 5. Table 4.1 shows how many of those 86 appear within the first 25, 50 and 100 positions in each final list.

Table 4.1: Discovery results for the 1,031-source labelled evaluation subset. The rank columns count sources scored 4 or 5 within each part of the ranked list. Discovery AUC summarises the full ranking.

Feature extractor	Discovery AUC	Rank 25	Rank 50	Rank 100
BYOL	0.784	10	16	28
DINOv3 ViT-H+/16	0.928	24	45	65

DINOv3 placed sources scored 4 or 5 earlier. Its discovery AUC exceeded BYOL's by 0.144. By rank 100 it had recovered 65 of the 86 high-interest sources (75.6%), compared with 28 (32.6%) for BYOL, a difference of 37 sources. The larger AUC shows that the difference was not confined to this one rank.

Figure 4.1 shows this progression. A curve that rises sooner indicates that more interesting sources would be encountered early. The dashed diagonal represents a random inspection order, not a separate model run. The table and curves report the same rankings in different forms.

The saved discovery arrays reproduce the counts and AUC values. Repeated runs gave the same point estimates. No uncertainty interval was calculated, so this repeatability does not measure how the ranking would vary on an independent set of sources.

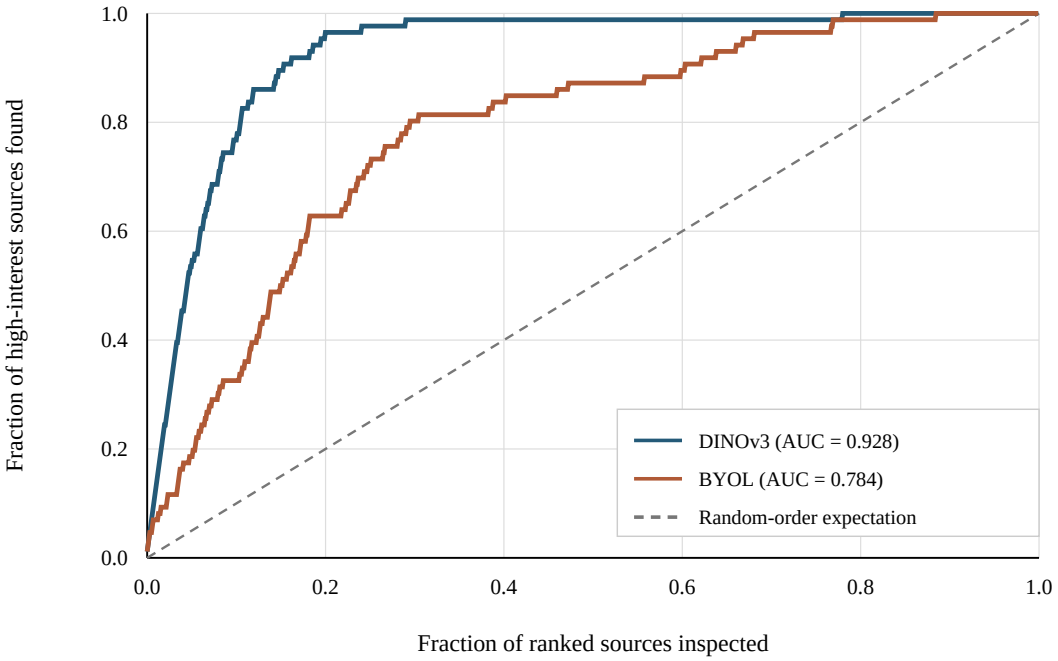


Figure 4.1: Cumulative discovery of the 86 sources scored 4 or 5 among the 1,031 labelled evaluation sources. The curves use the saved BYOL and DINOv3 discovery arrays. The dashed diagonal shows a random inspection order.

4.2 Highest-ranked radio sources

The top 20 radio cutouts from each ranking are shown in figs. 4.2 and 4.3. The accompanying Top-20 tables identify 20 sources scored 4 or 5 in the DINOv3 list and eight in the BYOL list. An asinh stretch was applied separately to each cutout for display, so apparent brightness cannot be compared quantitatively between them.

Only three source identifiers occur in both saved Top-20 tables. Their positions are given in table 4.2. Each has a human-interest score of 5.

Table 4.2: Sources shared by the saved BYOL and DINOv3 Top-20 rankings of the 1,031 labelled evaluation sources. Rank 1 is the highest predicted interest score.

Source identifier	BYOL rank	DINOv3 rank	Human score
`img28_isl2316`	7	10	5
`img30_isl3595`	14	4	5
`img39_isl2935`	8	3	5

The shared cutouts illustrate some of the structure visible in the high-ranked sources. `img39_isl2935` has a curved bright pattern with a brighter knot, while `img30_isl3595` shows several bright regions within more extended low-level emission. In `img28_isl2316`, the brighter regions form an approximately vertical grouping. These are visual descriptions of the displayed radio cutouts, not identifications of jets, lobes or distinct physical sources.

4.2.1 Illustrative sources unique to each ranking

Figure 4.4 shows three sources from each Top-20 list that do not appear in the other. All six have human-interest scores of 4 or 5. The enlarged cutouts make their visible structures easier to compare.

In the BYOL-only examples, `img79_isl931` has a compact bright grouping within fainter irregular emission. `img60_isl6079` shows two bright peaks connected by lower-level emission, while `img9_isl4406` has an incomplete curved bright outline and a spatially offset bright spot. Among the DINOv3-only examples, `img39_isl2421` shows two brighter regions joined by a narrow feature, `img65_isl986` contains a bent narrow bright structure, and `img60_isl2390` has broad patchy emission with several bright peaks.

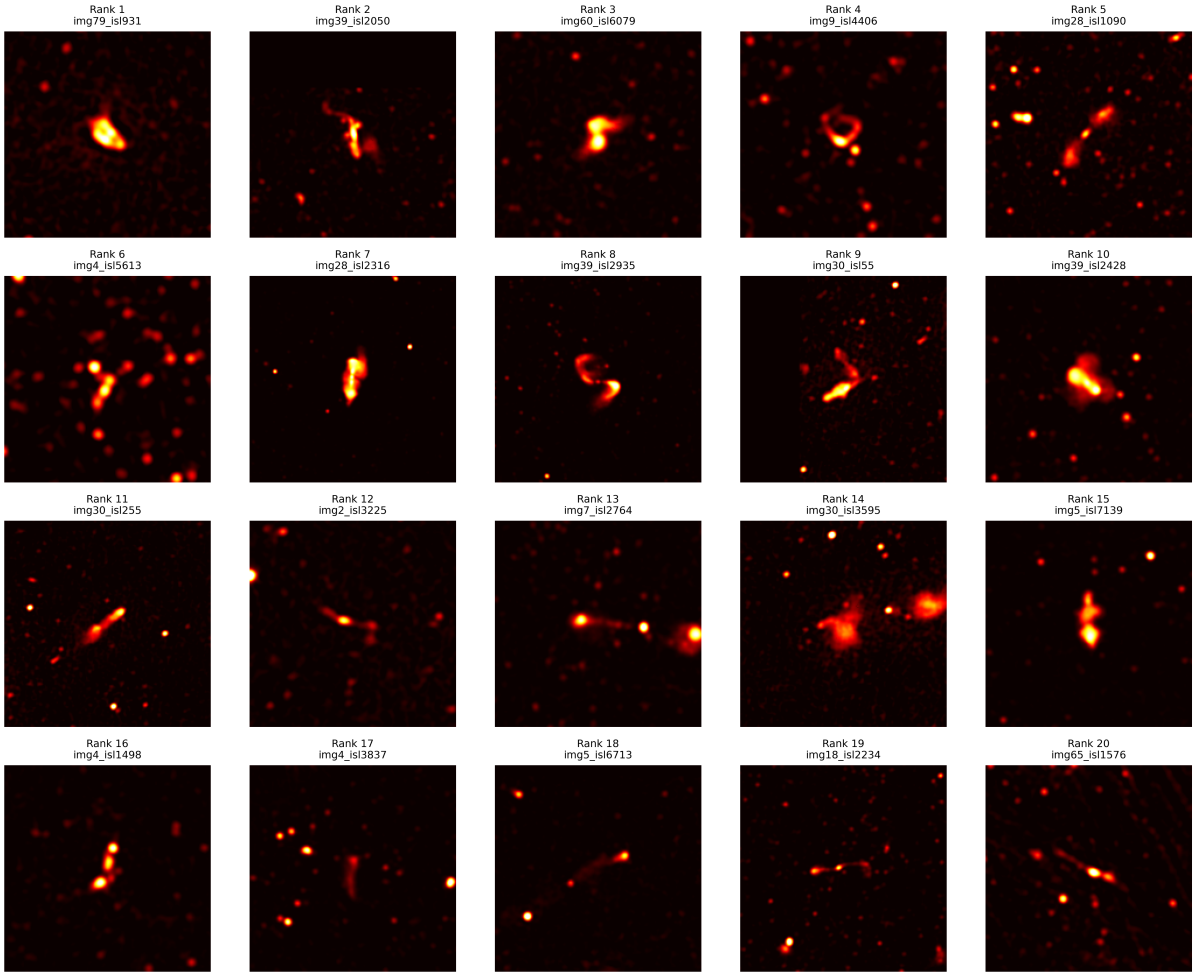


Figure 4.2: Top-20 MGCLS cutouts from the 1,031-source labelled subset, ranked by the BYOL-based Protege procedure. An asinh stretch was applied for display only. Image panel produced during this study.

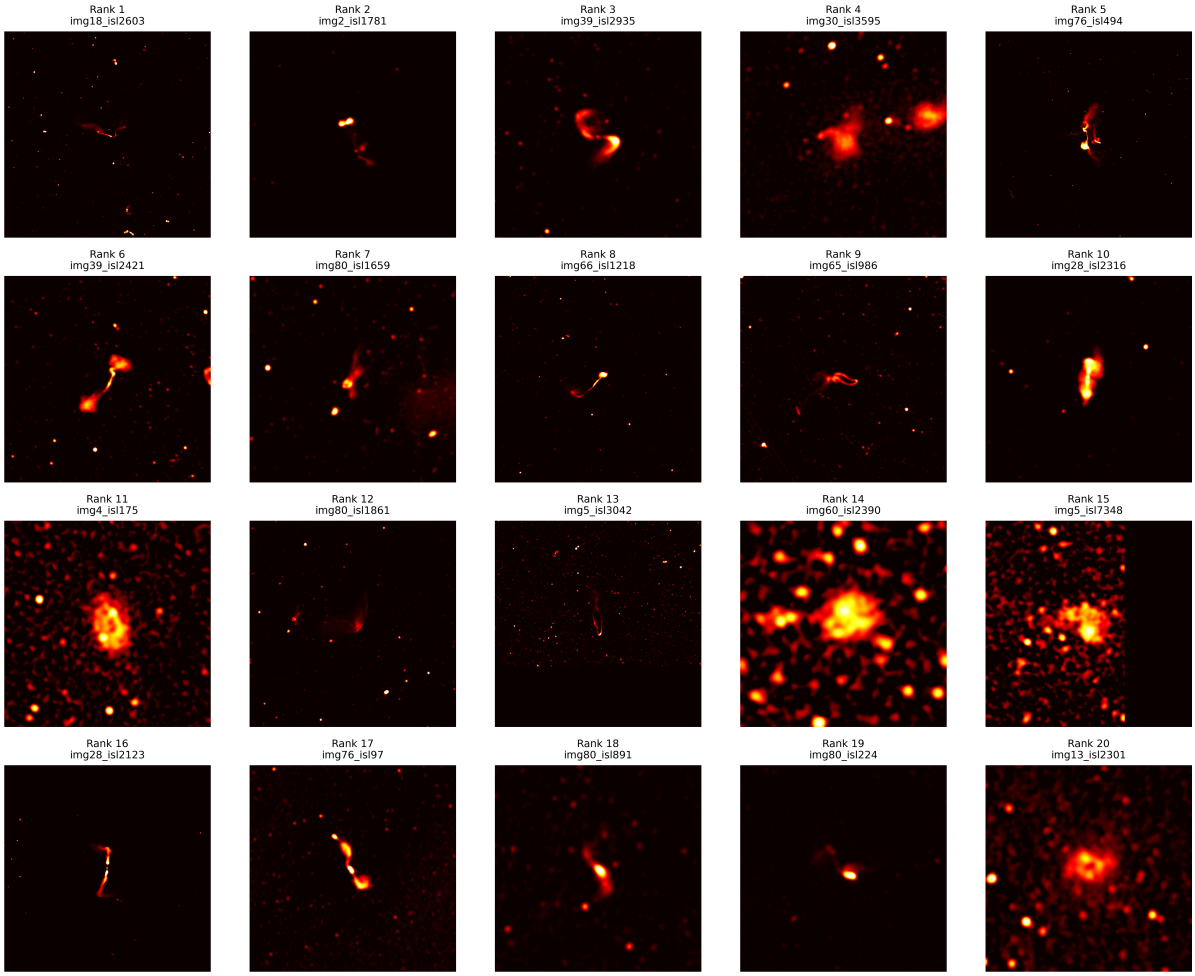


Figure 4.3: Top-20 MGCLS cutouts from the 1,031-source labelled subset, ranked by the DINOv3-based Protege procedure. An asinh stretch was applied for display only. Image panel produced during this study.

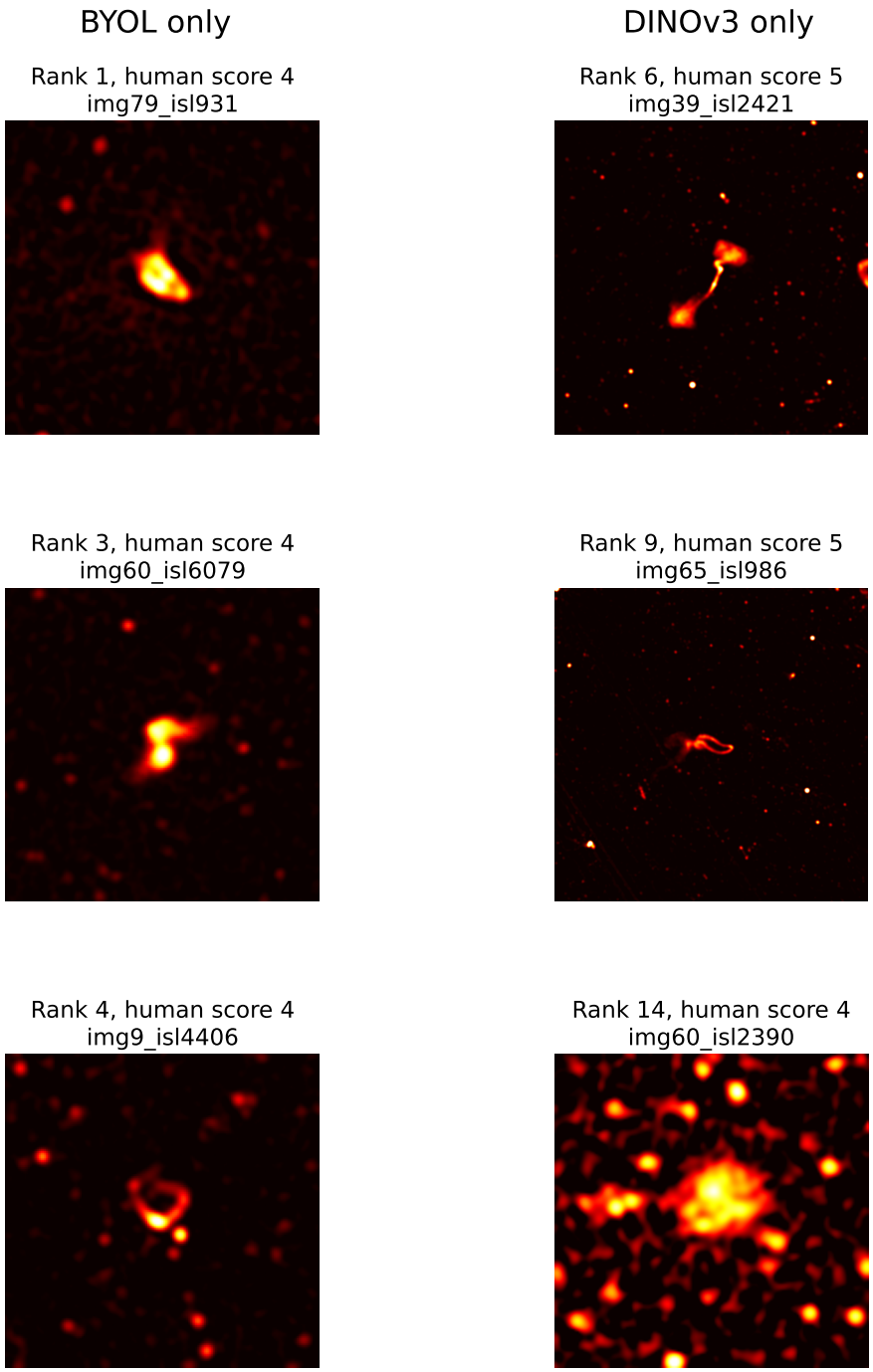


Figure 4.4: Six MGCLS sources occurring in only one saved Top-20 list. The left column contains BYOL-only sources and the right column contains DINOv3-only sources. Ranks and human-interest scores are from the saved Top-20 tables. The images were cropped from the asinh-stretched panels in figs. 4.2 and 4.3, which were generated from the original MGCLS FITS images. The adaptive cutout size and per-image intensity stretch vary between sources, so apparent extent and brightness cannot be compared quantitatively. Panel prepared for this study.

5 Discussion

The two feature sets gave Protege different rankings of the same labelled sources. DINOv3 placed 37 more sources scored 4 or 5 within the first 100 recommendations than BYOL. The discovery curves show that the difference extended beyond that cut-off. Both feature sets went through the same PCA variance criterion and Protege settings. The different extracted features led to different recommendation lists.

Protege's GP uses the PCA-reduced feature vectors rather than the images themselves. Different embeddings change the distances between sources that the GP uses when predicting interest. Those distances were not analysed here, so the results do not explain why DINOv3 ranked particular sources higher.

Only three sources appeared in both Top-20 lists. The six larger cutouts in fig. 4.4 show compact groupings, curved narrow features and broader patchy emission on both sides. These selected examples do not establish how often any structure occurs in the full rankings. A consistent morphology assessment, supported by wider radio or multiwavelength images where needed, would be required to compare physical source types.

Earlier Protege studies used human feedback to find overlooked MGCLS sources [Lochner and Rudnick, 2025] and to prioritise diffuse emission [Etsebeth et al., 2026]. Their objectives and samples differ, so their numerical results are not direct benchmarks here.

DINOv3's higher score does not show that a pretrained vision model will always outperform a radio-trained one. The 95% PCA threshold retained 45 BYOL components and 195 DINOv3 components, so Protege received feature spaces with different dimensions. The two extractors also differed in training and input normalisation. The ranking difference cannot be attributed to model architecture alone.

The reported rankings include sources whose recorded scores were supplied to Protege during training. The discovery values therefore describe this labelled sample, not performance on unseen sources. A separate test set is needed to assess that.

6 Conclusion

The pretrained DINOv3 ViT-H+/16 extractor was integrated with Protege, and its ranking was compared with the supplied BYOL baseline using discovery metrics and the highest-ranked cutouts. DINOv3 reached a higher discovery AUC, 0.928 compared with 0.784, and brought more high-interest sources into the first 100 recommendations. Only three sources appeared in both Top-20 lists. For this labelled sample, DINOv3 was the more effective of the two representations for prioritising sources for inspection.

The evaluation included sources whose scores had been used during Protege training, so the result does not establish performance on unseen sources. The selected cutouts are also insufficient to determine whether either method favours a physical morphology. Those questions require withheld labels and a wider image assessment.

7 Future work

The first priority is an evaluation in which a set of human-interest labels is withheld from Protege throughout active learning. Both representations should be ranked against that same held-out set, with the label budget, initial selection and acquisition settings fixed in advance. Repeated runs with recorded random seeds would show how sensitive the rankings are to the choice of initial labelled sources. Future runs should also save separate feature, PCA and final-ranking files for each model so one run cannot overwrite another's outputs.

A controlled input comparison could test whether preprocessing contributes to the ranking difference. The same radio cutouts could be processed with matched window sizes, scaling and background treatment. Any input requirements that cannot be matched between the models should be recorded. This matters especially for faint extended emission, which can change visibly under different intensity transforms. The global DINOv3 vector could also be compared with a defined patch-level summary, with the Protege settings kept unchanged.

Finally, the highest-ranked candidates could be assessed using a consistent morphology scheme and wider radio or multiwavelength images. Testing on an additional survey would show whether the ranking difference extends beyond this MGCLS sample.

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A Code and accompanying files

The selected extracts below show the feature-extraction, evaluation and ranking steps used in this comparison. The complete notebooks and feature-extraction files accompany the report.

A.1 Feature extraction

Professor Michelle Lochner supplied the completed BYOL features and the BYOL baseline workflow. The supplied feature-extraction module is retained as `code/feature_extraction/byol_features.py`. Its batched extraction step is shown in code A.1. This project adapted that workflow to use a pretrained DINOv3 encoder. The DINOv3 module is retained as `code/feature_extraction/dinov3_features.py`. Codes A.2 and A.3 show the input normalisation and inference steps. The full module also loads the pretrained model from local files. Those paths refer to the computer on which extraction was run and must be changed before running it elsewhere.

Code A.1: Batched feature extraction in the supplied BYOL module.

```
        all_feats = []
        self.learner.eval()

        t1 = time.perf_counter()

        with torch.no_grad():
            for i, image_batch in enumerate(data_loader):
                image_batch = image_batch[0]
                images = image_batch.to(self.device)

                # Run model on batch
                _, embedding = self.learner(images, return_embedding=True)

                # Convert to numpy and collect
                feats = embedding.detach().cpu().numpy()
```

Code A.2: DINOv3 input normalisation.

```
        if normalize:
            # Normalisation uses the channel statistics of the pretrained model.
            transform_list += [transforms.Normalize(
                mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])]
        self.transforms = transforms.Compose(transform_list)
```

Code A.3: DINOv3 inference without gradient updates.

```
        all_feats = []
        self.model.eval()

        t1 = time.perf_counter()

        with torch.no_grad():
            for i, image_batch in enumerate(data_loader):
                image_batch = image_batch[0]
                images = image_batch.to(self.device)

                # The pretrained model returns an embedding for each image.
                with torch.no_grad():
                     embedding = self.model(images)


                # Each embedding is stored as a Numpy array for the output table.
                feats = embedding.detach().cpu().numpy()
                all_feats.append(feats)
```

A.2 Evaluation and Top-20 panels

The BYOL and DINOv3 notebooks are `code/notebooks/mgcls_automated_example.ipynb` and `code/notebooks/mgcls_-DINOv3.ipynb`. They use the same labelled subset and Protege settings. Code A.4 shows the saved DINOv3 notebook's call to the existing active-learning procedure and the rank-count check. The BYOL notebook uses the corresponding call. The two notebooks then save the first 20 sources in descending order of the final trained score, as shown in codes A.5 and A.6.

Code A.4: Protege evaluation and rank counts in the DINOv3 notebook.

```
active_output = run_GP_active_learning(
    features_eval,
    labels,
    anomaly_scores,
    output_dir,
    steps=10,
    initial_steps=10,
    N_labels=600,
    ei_tradeoff=0,
)

# %% Notebook cell 16
# The final Protege score gives the order in which sources would be inspected.
ranked_ids = active_output.sort_values(
    "trained_score", ascending=False
).index

ranked_labels = labels.loc[ranked_ids, "binary_label"].astype(int)

# Each count is the number of high-interest sources found by that rank.
for rank in (25, 50, 100):
    found = int(ranked_labels.iloc[:rank].sum())
    print(f"First {rank} sources inspected: {found} high-interest sources found")
```

Code A.5: Saving the BYOL Top-20 table.

```
# These are the source IDs in the first 20 positions of the final ranking.
top_20_ids = active_output.sort_values('trained_score',
    ascending=False).head(20).index

# The accompanying table keeps the scores and labels for those sources.
top_20_info = (active_output.sort_values("trained_score", ascending=False).head(20))

# Resetting the index keeps each source ID in the saved CSV.
top_20_info = top_20_info.reset_index()

# These columns show the ranking information used in the comparison.
top_20_info[["index", "score", "human_label", "trained_score", "acquisition"]]

top_20_info.to_csv("BYOL_PROTEGE_Top20_info.csv", index=False)
```

Code A.6: Saving the DINOv3 Top-20 table.

```
# These are the source IDs in the first 20 positions of the final ranking.
top_20_ids = active_output.sort_values('trained_score', ascending=False).head(20).
    index

# The accompanying table keeps the scores and labels for those sources.
top_20_info = (active_output.sort_values("trained_score", ascending=False).head(20))

# Resetting the index keeps each source ID in the saved CSV.
top_20_info = top_20_info.reset_index()

# These columns show the ranking information used in the comparison.
top_20_info[["index", "score", "human_label", "trained_score", "acquisition"]]

top_20_info.to_csv("DINOv3_PROTEGE_Top20_info.csv", index=False)
```

Code A.7: Image display code for the Top-20 panels.

```
def asinh_display(img, stretch=5.0):
    # Non-finite FITS values become background pixels in the display.
    image = np.nan_to_num(np.asarray(img, dtype=float), nan=0.0, posinf=0.0,
    neginf=0.0)
    # Negative values are hidden, while zero-padded edges stay black.
    image = np.clip(image, 0, None)

    positive = image[image > 0]
    # A cutout with no positive pixels has nothing to stretch.
    if positive.size == 0:
        return np.zeros_like(image)

    # The high percentile stops a few bright pixels setting the whole scale.
    upper = np.percentile(positive, 99.9)
    scaled = np.clip(image / upper, 0, 1)
    # Asinh brings out faint emission while compressing the bright end.
    return np.arcsinh(stretch * scaled) / np.arcsinh(stretch)

# This affects displayed cutouts only, the feature input is unchanged.
image_dataset_full.display_transform_function = asinh_display
```

A.3 Saved records and limits of the package

The BYOL, DINOv3 and discovery-curve notebooks are included with the accompanying files.

The original MGCLS FITS images and the pretrained DINOv3 checkpoint are not included because of their size. Re-extracting features therefore requires those external files and a compatible Astronomaly and DINOv3 installation. The saved features and evaluation records allow the comparison in this report to be checked without re-extracting them.